

Ambient Gas Sensor Candidates

These four sensors are candidates for ambient gas detection alongside the existing inline sensor suite. Unlike pressure and flow sensors, they measure actual gas composition in the air near pipeline joints and hospital rooms, providing a secondary leak detection layer.

1. Gravity: Electrochemical Oxygen / O₂ Sensor (DFRobot, SEN0322)

Measures ambient O₂ concentration by detecting the electrochemical oxidation current at the working electrode. Relevant because oxygen leaks create both patient safety and fire hazards that pressure sensors cannot detect. Outputs a raw analog current requiring an external transimpedance amplifier and ADC, not I²C. Has a 2-year lifespan and requires periodic recalibration due to monthly drift.

Parameter	Value
Detection Gas	O ₂
Measurement Range	0–25% Vol (max 30% Vol)
Response Time (T ₉₀)	≤ 15 s
Interface	Analog current output (0.08–0.25 mA in air); requires ADC
Stability / Drift	< 2% per month; zero drift ≤ 1% Vol (–20°C to 40°C)
Operating Temp / Humidity	–20°C to +50°C; 0–99% RH non-condensing
Form Factor	Φ20 mm cylindrical; 16.6 mm height
Lifespan	2 years

Integration note: Cannot be encapsulated in resin or stored in O₂-free environments. Electrodes must remain open-circuit during storage.

2. Sensirion SGP41 - VOC and NOx Air Quality Sensor

Metal oxide (MOx) sensor providing a VOC Index and NOx Index (both 1–500 scale) over I2C. Intended for detecting anesthetic gas near pipeline joints, but does not give true N2O ppm, only a relative index. High false-positive risk in hospitals due to MOx sensitivity to cleaning agents and hand sanitizers. Easiest to integrate: follows the same Sensirion I2C pattern as STS41A and SDP33.

Parameter	Value
Detection Modalities	VOC Index + NOx Index (1–500 scale each)
True Gas Concentration?	No - relative index only, not ppm
NOx Limit of Detection	< 20 ppb NO2 equivalent
Response Time	VOC: < 10 s; NOx: < 250 s
Interface	I2C, address 0x59, up to 400 kHz
Power	3.0 mA @ 3.3 V (during measurement)
Package	2.44 × 2.44 × 0.85 mm DFN
NOx Stabilization Time	Up to 7 days continuous operation for full NOx spec
Operating Temp / Humidity	–10°C to +50°C; 0–90% RH non-condensing

Integration note: Requires 10s conditioning after power-up. Sensirion Gas Index Algorithm must run on the main CPU. Humidity compensation can be fed from the existing onboard humidity sensor. Avoid power-cycling if NOx tracking is needed.

3. Dynament Platinum Series - Nitrous Oxide NDIR Sensor

NDIR sensor giving true N₂O concentration in ppm over UART. This is the most medically credible N₂O detection option reviewed. Used in healthcare wearable N₂O detectors worn by OR staff. Resolution of 20 ppm is marginal for the 25 ppm NIOSH occupational limit. Significantly higher power draw and larger form factor than chip-level alternatives. Requires UART peripheral rather than I²C, and a dedicated PCB socket.

Parameter	Value
Detection Gas	Nitrous Oxide (N ₂ O)
Measurement Range / Resolution	0–1,000 ppm at 20 ppm resolution; or 0–1% volume
True Gas Concentration?	Yes - linearized, temperature-compensated ppm output
Accuracy	±5% at 20°C and 1 bar with calibration gas
Response Time (T ₅₀ / T ₉₀)	< 15 s / < 30 s
Long-Term Zero Drift	±5% of full scale per month (≈ ±50 ppm on 0–1000 ppm range)
CO ₂ Cross-Sensitivity	1,000 ppm CO ₂ = < 100 ppm N ₂ O equivalent
Warm-Up Time	~10 minutes to full stability
Interface	UART, 2.8 V logic, 38,400 baud default (not I ² C)
Power	80 mA at 3 V (240 mW)
Form Factor	Φ20 mm cylindrical, 16.6 mm height, 15 g
MTBF / Certifications	> 5 years; Industrial Ex d IIC, Mining M1, UL Approved

Integration note: Requires a dedicated PCB socket (Wearnes Cambion 450-3326-01-06-00). Do not solder directly to pins. 2.8 V logic level, so might want to include a voltage divider for the TX pin which is 3.3V. Plan for monthly recalibration in long-term deployment.

4. Sensirion SCD41 - Photoacoustic CO2 Sensor

Optical CO2 sensor using photoacoustic NDIR with an integrated temperature and humidity sensor for on-chip compensation. Provides true CO2 concentration in ppm over I2C. Relevant for CO2 pipeline monitoring (used in laparoscopic surgery); high ambient CO2 poses asphyxiation risk. Single-shot mode allows very low average current (0.45 mA) for battery-conscious nodes. Automatic self-calibration (ASC) makes it maintenance-free under normal conditions.

Parameter	Value
Detection Gas	CO2 (plus integrated RH and temperature)
Accuracy (400–1,000 ppm)	±(50 ppm + 2.5% of reading)
Accuracy (1,001–5,000 ppm)	±(50 ppm + 3–5% of reading)
True Gas Concentration?	Yes - direct ppm output
CO2 Response Time (τ_{63})	60 s
Interface	I2C, address 0x62, up to 400 kHz
Power (periodic, 1/5 s)	15–18 mA @ 3.3 V; peak up to 205 mA
Power (single shot, 1/5 min)	0.45 mA average @ 3.3 V
Package	10.1 × 10.1 × 6.5 mm LGA
Calibration	Automatic self-calibration (ASC); maintenance-free with weekly ~400 ppm CO2 exposure
Lifespan	> 10 years (simulated)
Operating Temp / Humidity	–10°C to +60°C; 0–95% RH non-condensing

Integration note: Peak current of 205 mA requires a low-noise LDO regulator. Keep-free area around thermal relief hole on PCB footprint must be respected. CO2 accuracy is only specified above 400 ppm. Accuracy fully restores within 5 days after reflow soldering.

Summary Comparison

Property	SEN0322	SGP41	Dynament Platinum	SCD41
Target Gas	O ₂	VOC / NO _x	N ₂ O	CO ₂
Interface	Analog (ADC req.)	I ² C	UART	I ² C
True ppm Output?	Yes (% Vol)	No (index)	Yes (ppm)	Yes (ppm)
Power Draw	Low	3 mA	80 mA	0.45–18 mA
Form Factor	Φ20 mm cylinder	2.44×2.44 mm chip	Φ20 mm cylinder	10.1×10.1 mm LGA
Warm-up	Fast	10 s / 7 days (NO _x)	~10 min	Fast
Recalibration	Yes (monthly)	No (auto)	Yes (monthly)	No (ASC)
Interferent Risk	Low	High (sanitizers)	Low	Low
Lifespan	2 years	10 yr	> 5 yr	> 10 yr